



UTM
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FACULTY OF COMPUTING

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ENTERPRISE SYSTEM & DESIGN MODELLING

ASSIGNMENT 1 - TECHNICAL REPORT

AUTOMATING REPETITIVE BUSINESS PROCESSES
TO IMPROVE EFFICIENCY AND REDUCE ERRORS

SECTION : 01

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Executive Summary

This report examines how the organizations contain the Robotic Process Automation (RPA) in their Enterprise Resource Planning (ERP) systems to enhance operational performance. Many ERP cores still rely on manual processes and rule-based tasks, which can delay the process and output, create mistakes and increase burden on employees with the overload tasks.

The analysis and discussion shows that applying RPA to functions like invoice management and purchase order updates are faster and can reduce the errors. A real example reported that there were improvements in processing time and compliance when the automation tools appeared. However, findings also highlight that automation cannot simply be used on inefficient processes, as this can create more dysfunctional activities. It must be carefully handled and planned to avoid any disruption functions.

For having better outcomes, the report suggests some methods for instance strong governance operations, continuous monitoring and ongoing workforce development. These strategies help organizations in strengthening their productivity and advancing their digital transformation goals.

Introduction

As our modern lifestyle gets more and more digitized, businesses are engaging automation technologies in their functions to enhance efficiency, accuracy, and scalability. One of the groundbreaking innovations is Robotic Process Automation (RPA) – a computer software for automating monotonous rule-based

tasks completed by human beings typically (Aguirre & Rodriguez, 2017). RPA accelerates information processing and reduces human errors by applying "software robots" that imitate human involvement with computer programs like Enterprise Resource Planning (ERP) systems. Besides that, RPA also enables workers to focus on strategic value-added activities while smooths out the productivity by decreasing manual intervention.

One of the significant achievements is the incorporation of RPA into the ERP systems in modern enterprises digitalization. ERP systems are the core of finance, procurement, and supply chain activity management yet involve weary tasks such as data entry, verification, and reconciliation across systems. RPA outlines these challenges, improves process consistency and accuracy of operations through the automation of tasks like data migration, report creation, and invoice processing (Willcocks et al., 2019). Prioritizing a larger efficiency, transparency, and decision-making in real time based on Industry 4.0 goals is made possible through this combination of automation and ERP processes.

Enterprises that applied RPA have attained outstanding improvements in their operational performance. The joint ERP modernization report by UiPath and Deloitte depicts it as a transformation through automation-led ERP integration that can be measured in terms of precision, effectiveness, and compliance monitoring (UiPath Inc., n.d.). Thus, these results together not only show the measurable benefits of the automation of repetitive business processes in company systems

but also underline RPA's role in the digital transformation of enterprises.

The primary goal of this study is to investigate the role of RPA in increasing productivity and decreasing human errors in ERP systems.

Background/Literature Review

Robotic Process Automation (RPA) has appeared as a leading automation method within enterprise systems, especially in supporting repetitive and rules-based tasks. RPA technology uses customizable software bots that can replicate the user actions on digital systems, including the data entry, validation and reporting (Syed et al., 2020). RPA was first adopted in finance, healthcare and public sectors and had been reported that RPA considerably enhances the process accuracy, improves compliance and reduces operational delays. RPA also minimizes the processing time by 80% and improves accuracy to 95-100% (*Robotic Process Automation Explained*, 2022). The quick adoption of RPA is conducted by the ERP platforms where it can provide strong support for automation tools.

This also highlights the development of RPA from the elementary task automation toward Intelligent Process Automation (IPA), which embraces Artificial Intelligence (AI), machine learning and process mining abilities (Siderska et al., 2023). The organizations can automate more complex processes like invoice verification, predictive analysis and supply the exception handling. When combined with the ERP systems, intelligent RPA also supports rapid decision making, real-time monitoring and can reduce dependency on

manual work operations (Flechsigg et al., 2021). Despite its benefits, researchers find some limitations. If the business process is disorganized and inconsistent, it also can lead to a messy manual process, automation failure and unintended process disruptions. RPA is already a good process but also enhances the dysfunctional of a bad one. The first step in any RPA implementation is crucial in a process and standardization. Organizations must also label the cybersecurity risks, system maintenance and sharpen the employee's skill to ensure long term success.

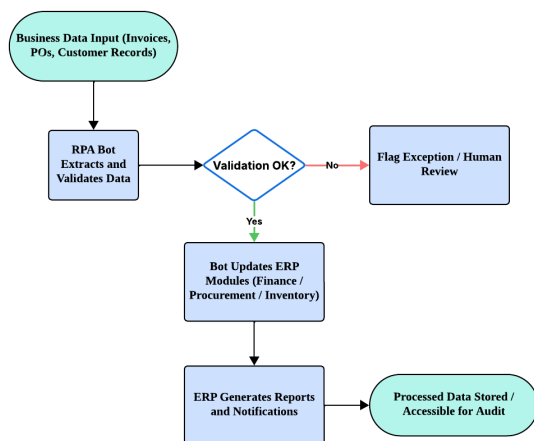
In general, current researchers agree that RPA can serve as an enhancer in digital transformation along with the future trend. This great combination of RPA, AI and advanced analytics can maximize the efficiency among the enterprise systems.

Analysis and Discussion

After the increasing application and adoption of RPA technologies in business systems as pointed out before, let's dive deeper into the use of RPA in ERP systems, its practical application, and its effect on efficiency and error reduction. Robotic Process Automation (RPA) operates using the 'software robots' that replicate human interactions with enterprise systems through the user interface or Application Program Interfaces (APIs). The bots are designed to tackle structured, rule-based activities like data migration, transaction processing, and report generation in Enterprise Resource Planning (ERP) settings (Aguirre & Rodríguez, 2017). RPA is exercised through connectors or custom scripts to automate repetitious tasks such as invoice verification, purchase order approval, and

data synchronization between modules by applying it within ERP solutions such as Systems, Applications, and Products (SAP), Oracle ERP, or Microsoft Dynamics (Willcocks et al., 2019).

Due to the fact that RPA assists in automating high-volume, repetitive tasks and lessening human interaction in advanced systems such as ERP, its enactment is outstretched rampantly (Costa et al., 2022) . This implies that the human intervention level in transactional processes is minimized, therefore efficiently enhancing while eliminating chances of human errors. RPA adaptability also offers the capability of simultaneously running many bots in departments, promising consistency and managing high volumes of information that would be difficult for human workers to accomplish manually.



RPA bots interacting with ERP modules

The collaboration between UiPath and Deloitte to automate processes in SAP ERP landscapes is one example of how RPA has proved to be effective in ERP modernization. The case study reveals how organizations and businesses implement RPA-based automation platforms to

achieve improved compliance tracking, greater data integrity, and reduced operational latencies (UiPath Inc., n.d.). Deloitte successfully replaced repetitive manual entries with autonomous bots that handled invoice processing, reconciliation, and reporting tasks, thanks to the UiPath's automation tool.

In the aspect of speed and accuracy, the result established quantifiable improvement together with the reduction of human error in purchase order management. In addition, it identified the strategic importance of automation-led ERP modernization—where firms move from manual tasks execution to invasive, intelligent automation that aids decision-making and process optimization (Măldăreanu, 2025).

Process Automated	Outcome	Impact on Efficiency
<i>Invoice Processing</i>	Automated end-to-end data extraction, validation, and posting in SAP FI module	↓ Manual effort by 40% ↑ Speed of approval cycle
<i>Purchase Order Management</i>	Real-time compliance and matching using rule-based bots	↓ Error rate by 25% ↑ On-time PO closure
<i>Financial Reconciliation and Reporting</i>	Bots consolidate ledgers and generate reports automatically	↓ Processing time by 30% ↑ Data accuracy to 98%

Summarize of Case Highlights

Reducing cycle times and accelerating transaction processing are examples of numerous advantages when it comes to RPA application in ERP environments. Accuracy and adherence are enhanced while the inconsistency problem that roots from human labor is eliminated since they follow strict pre-coded rules. RPA is also cost-effective and enables human employees to focus on more superior business analysis and decision-making functions (Willcocks et al., 2019; Costa et al., 2022).

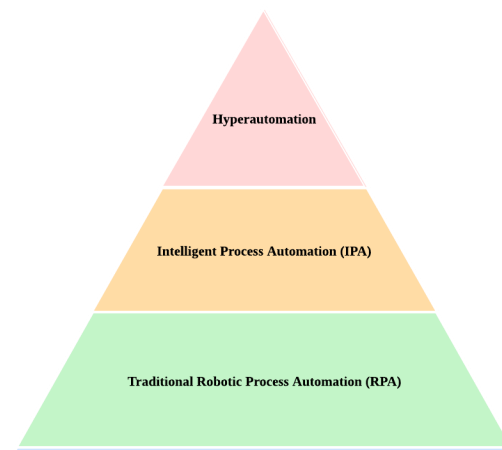
Well, there are some challenges and limitations. Especially for smaller and medium enterprises that require onetime framework and integration support, the implementation cost might be high. There is also defiance risk from employees as they might feel that their jobs are under threat. On a business front, RPA solutions can be affected if ERP interfaces or flows are modified, leading to regular updates and bot refresh.

Based on (Siderska et al. 2023), the large part of RPA success relies on the standard of fundamental business processes. Clustered workflows can lead to automation failure or promote existing inefficiencies instead of getting rid of them. Unlike other enterprise software such as Customer Relationship Management (CRM) or Human Resource Management (HRM) systems, ERP automation is far more complex due to the extensive interconnectivity of its modules.

Anyway, RPA integration within ERP systems is always troubled by integration complexities, data security, and scalability. Furthermore, sturdy cybersecurity

practices and audit rules need to be implemented to prevent them from leaking or being accessed illegally since the bots process sensitive operational and financial information (Costa et al., 2022).

Somewhere down the line, RPA is on its way to hyperautomation—a de facto name for Intelligent Process Automation (IPA) as already described—where automation is complemented by Artificial Intelligence (AI), machine learning, and process mining to develop intelligent, adapting systems. The next stage will enable bots not just to apply rules but also decide in real time from data, further grounding ERP savvy and flexibility.



Progression of ERP Automation

Recommendations

Based on the analysis and discussion, implementing the RPA in ERP systems should consider the following strategies for having the successful implementation of digital transformation:

Handle Process Assessment Before Proceed with Automation

The organization should identify high volume and rule-based processes to maximize the return on investment. Process mining techniques are used to

support the accurate task (Wewerka & Reichert, 2020).

Adopt a Structured Governance Model

Ensuring the clear guidelines for bot ownership, cybersecurity and maintenance are important to avoid system disruption. This also ensures agreement with the industry standards and can be reused when there are any repeatable RPA steps (Kokina et al., 2024).

Combine the RPA with AI to Enable Intelligent Process Automation (IPA)

Collaborating the RPA with machine learning and predictive analytics can expand the automation over the basic tasks into conceptual workflows like fraud detection and decision support. AI also can help interpreting and improving the robustness of RPA solutions (Wewerka & Reichert, 2020).

Measure and Distribute The Performance Continuously

The Key Performance Indicators (KPI) like error rate reduction and process cycle time should be monitored to ensure the improvement, customer satisfaction and employee engagement (Huda et al., 2024).

Invest More in Workforce Upskilling and Change Management

Employees must be trained for the roles involving the bot monitoring, exception handling and automation design. This could help to reduce the resistance and ensure that employees remain competitive in the AI-driven job market (Benjamin & Matthew, 2025).

By applying all strategies, the organizations can fully maximize the operational effectiveness, reduce human error and achieve the sustainable digital transformation.

Conclusion

In essence, this paper has examined how Robotic Process Automation (RPA) reduces human fault and improves efficiency in Enterprise Resource Planning (ERP) systems. The discussion affirms that RPA can systematise repetitious, rule-based processes, increase data authenticity, and allow quicker decision-making in enterprise processes if implemented properly. The UiPath and Deloitte SAP ERP modernization project is a real-world application that showcases the technology's actual impact on accordance, cost reduction, and process performance.

Other than that, it is also emphasized that automation succeeds if the workforce have standardized processes, resolute governance, and continuous upskilling. As stated before, the combination of RPA with Artificial Intelligence (AI) and analytics — referred to as Intelligent Process Automation (IPA) — is the next stage of enterprise digital transformation. In the future, enterprises must continue to invest wisely in secure, scalable, and intelligent automation platforms that complement human intelligence. Through this balance of people and technology, it will ensure sustainable efficiency, accuracy, and innovation in modern enterprise platforms.

Reference

- Aguirre, S., & Rodriguez, A. (2017). Automation of a business process using Robotic Process Automation (RPA): a case study. In *Communications in computer and information science* (pp. 65–71). https://doi.org/10.1007/978-3-319-66963-2_7
- Benjamin, & Matthew. (2025). *The Impact of AI and RPA on Workforce Transformation*. ResearchGate.
https://www.researchgate.net/publication/389868494_The_Impact_of_AI_and_RPA_on_Workforce_Transformation
- Costa, D. A. S., De Gusmão, A. P. H., & Da Silva, L. C. (2022). *Robotic Process Automation (RPA) Adoption: A Systematic Literature Review. Engineering Management in Production and Services*, 14(2), 81–95. <https://doi.org/10.2478/emj-2022-0012>
- Flehsig, C., Anslinger, F., & Lasch, R. (2021). *Robotic Process Automation in purchasing and supply management: A multiple case study on potentials*. ScienceDirect. <https://www.sciencedirect.com/science/article/pii/S1478409221000522>
- Huda, M., Rahayu, A., Furqon, C., Adib Sultan, M., & Susi Susilawati Sugiana, N. (2024). Analyzing the Impact of Robotic Process Automation (RPA) on Productivity and Firm Performance in the Service Sector. *(IJACSA) International Journal of Advanced Computer Science and Applications*, 16. https://thesai.org/Downloads/Volume16No6/Paper_35-Analyzing_the_Impact_of_Robotic_Process_Automation.pdf
- Kokina, J., Langmann, C., Kogan, G., & Weiser, M. (2024). Navigating the RPA Governance Landscape: Evidence from the Field. *Navigating the RPA Governance Landscape: Evidence From the Field*. <https://doi.org/10.2139/ssrn.5029201>
- Măldăreanu, A. (2025). *The Integration of ERP Systems and Technologies for Process Automation: A Bibliometric Analysis. Proceedings of the International Conference on Business Excellence*, 19(1), 341–354. <https://doi.org/10.2478/picbe-2025-0029>

- Robotic Process Automation explained. (2022, February 1). Deloitte. <https://www.deloitte.com/mt/en/services/consulting/perspectives/robotic-process-automation-explained.html>
- Siderska, J., Jadaan, K., & Pawlewski, P. (2023). *Intelligent Process Automation in Practice: A Review of Current Trends and Future Prospects*. *Procedia Computer Science*, 217, 214–223. <https://doi.org/10.1016/j.procs.2022.12.240>
- Siderska, & Suesse, J., & Aunimo, T., & Kedziora, L., & Von Stamm, D., & Aini, J., & Suraya. (2023). *Towards Intelligent Automation (IA): Literature Review on the Evolution of Robotic Process Automation (RPA)*. ResearchGate. https://www.researchgate.net/publication/376857611_Towards_Intelligent_Automation_IA_literature_review_on_the_evolution_of_Robotic_Process_Automation_RPA_its_challenges_and_future_trends/citation/download?tp=eyJjb250ZXh0Ijp7InBhZ2UiOiJwdWJsaWNhdGlvbGlzInByZXZpb3VzUGFnZSI6bnVsbH19
- Syed, R., Suriadi, S., Adams, M., Bandara, W., J.J. Leemans, S., Ouyang, C., H.M Ter Hofstede, A., Van De Weerd, I., Thandar Wynn, M., & A. Reijers, H. (2020). *Robotic Process Automation: Contemporary themes and challenges*. ScienceDirect. <https://www.sciencedirect.com/science/article/abs/pii/S0166361519304609>
- UiPath Inc. (n.d.). *Case Study: UIPAtH and Deloitte Revolutionize ERP Modernization | UIPATH*. UiPath and Deloitte: Revolutionizing ERP Modernization. A Blueprint for an Agentic Automation-led Approach. <https://www.uipath.com/resources/automation-case-studies/uipath-and-deloitte-revolutionizing-erp-modernization>
- Wewerka, J., & Reichert, M. (2020, December 22). *Robotic Process Automation -- A systematic literature review and assessment framework*. arXiv.org. <https://arxiv.org/abs/2012.11951>
- Willcocks, L., Lacity, M., & Craig, A. (2019). *Robotic Process Automation: Strategic Transformation Lever for Global Business Services? Journal of Information Technology Teaching Cases*.

Appendix

Logbook: <https://github.com/users/mukhriz/projects/15/views/1>

*Some of the information provided by the AI tools was reviewed and refined to ensure accuracy and alignment with these assignment requirements.